When Do Random Effects Models Reduce to Fixed-Effect Alternatives? Conditions for Equivalence in Longitudinal Clinical Trials

R.G. Thomas

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Contents

1 Introduction **1**
1.1 The practical problem . . . 1
1.2 Foundational results . . . 2
1.3 The Frost-Kenward-Fox observation . . . 2
1.4 Constrained longitudinal data analysis . . . 2
1.5 The role of summary statistics . . . 3
1.6 Modern MMRM practice . . . 3
1.7 Present study . . . 3

2 Methods **4**
2.1 ADEMP structure . . . 4
2.2 Setting 1: two-timepoint pre-post design . . . 4
2.3 Setting 2: three-timepoint random slopes . . . 5
2.4 Simulation procedure . . . 6

3 Results **6**
3.1 Setting 1: two-timepoint equivalence . . . 6
3.2 Setting 2: three-timepoint random slopes . . . 9

4 Discussion **9**
4.1 Limitations . . . 10

5 Future research **10**

6 References **11**
6.1 Morris et al. (2019) ADEMP Compliance . . . 11

1 Introduction

1.1 The practical problem

In clinical trials with repeated measurements, mixed models for repeated measures (MMRM) and random effects models have become the standard analytic approach[1,2]. These models accommodate within-subject correlation, handle incomplete data under the missing-at-random assumption, and provide flexible co-

variance structures. However, they impose a substantial interpretive burden on collaborators unfamiliar with the statistical machinery of variance components, restricted maximum likelihood estimation, and covariance parameterization.

A natural question arises: under what conditions does the random effects model reduce to, or closely approximate, a simpler fixed-effect analysis that would be more accessible to the statistical practitioner? If such conditions can be identified, the simpler model offers an intuitive bridge to the same inferences, sparing the analyst from invoking a complex apparatus when it is not strictly necessary.

1.2 Foundational results

The relationship between random effects and fixed-effect analyses has been explored from several angles.[3] showed that in a two-group pretest-posttest design, the repeated-measures ANOVA subsumes the difference-score t-test: the interaction F-statistic equals the square of the two-sample t-test on difference scores. The difference-score analysis requires fewer assumptions (homogeneous variance of differences and normally distributed errors) than the full repeated-measures model (equal variance-covariance matrices across groups). This algebraic embedding demonstrates that when the research question concerns only whether treatment groups differ in their pre-to-post change, the simpler analysis is not merely an approximation but is arithmetically identical to a component of the more elaborate model.

[4] introduced the general framework for random-effects models in longitudinal data that has since become ubiquitous. In their formulation, the random effects induce a structured variance-covariance matrix on the marginal distribution of the observed data. The key insight for our purposes is that when this marginal covariance is left unconstrained—as in the unstructured covariance MMRM—the random effects become a computational device rather than a substantive modeling choice.

1.3 The Frost-Kenward-Fox observation

[5] established a methodological framework for comparing conventional and run-in trial designs using linear mixed models. In their treatment of the two-timepoint case, they showed that the Laird-Ware random-intercept formulation treating baseline and follow-up as correlated outcomes (their equation 8) yields identical point estimates and very similar standard errors to the analysis of covariance model that conditions on baseline (their equation 6). This near-equivalence holds provided the variance-covariance matrix is unconstrained and $\sigma_{12} < \sigma_1^2$ and $\sigma_1^2 = \sigma_2^2$ (i.e., homogeneous marginal variances and positive but not perfect correlation). The result builds on[6], who showed that Rao's reduction of the[7] generalized MANOVA model yields estimators identical to those from an analysis of covariance.

1.4 Constrained longitudinal data analysis

[8] formalized the constrained longitudinal data analysis (cLDA), in which baseline is included in the response vector and the model constrains the two treatment groups to share a common baseline mean—a consequence of randomization.[9] compared cLDA to the longitudinal ANCOVA model and showed that (a) with

complete data, the treatment effect estimates are effectively identical but the cLDA standard errors are marginally smaller; and (b) with missing baseline data, cLDA is more efficient because it can borrow information from the observed outcome model to impute baseline values implicitly under the missing-at-random assumption.[10] derived sample-size formulae for cLDA, and[11] provided a formal efficiency comparison of cLDA versus longitudinal ANCOVA, quantifying the power gain as a function of the proportion of missing baselines.

1.5 The role of summary statistics

An alternative thread, developed by[12] and[13], advocates analyzing repeated measures through summary statistics—typically the mean post-baseline value or a fitted slope for each subject—followed by a between-group comparison of these summaries.[14] extended this approach, showing that summary-statistic methods can be viewed as special cases of the general mixed model when the within-subject covariance is compound symmetric.[15] provided an influential clarification of the debate between change-score analysis and ANCOVA, demonstrating that ANCOVA does not require baseline balance for unbiasedness and that the common slope assumption is testable and often reasonable.

1.6 Modern MMRM practice

In contemporary pharmaceutical practice, MMRM with unstructured covariance has become the de facto primary analysis for continuous outcomes[1,2].[16] showed that MMRM achieves power gains over complete-case ANCOVA only when time-by-covariate interactions are included; without these interactions, ANCOVA can be more powerful.[17] proposed augmented MMRM estimators that improve robustness against model misspecification.[18] provided the small-sample degrees-of-freedom correction that makes MMRM valid in trials with modest sample sizes, which is essential for the settings where the random-effects and fixed-effect models are most likely to be practically interchangeable.

Textbook treatments of these relationships can be found in[19],[20], and[21].[22] provided a recent tutorial contrasting conditional and marginal approaches to analyzing change in randomized trials.

1.7 Present study

The present report investigates, through analytic argument and Monte Carlo simulation, the conditions under which random effects models for two-group longitudinal designs reduce exactly or approximately to fixed-effect alternatives. We consider three canonical settings:

1. **Two-timepoint (pre-post) designs:** MMRM with unstructured 2×2 covariance versus ANCOVA with baseline as covariate.
2. **Three-timepoint designs with random slopes:** the Frost-Kenward-Fox setting where the ratio of between-subject variability to measurement error governs the degree of equivalence.
3. **Constrained longitudinal data analysis versus ANCOVA:** the Liang-Zeger formulation and its relationship to the conditional model.

For each setting, we identify the parameter configurations under which the simpler model provides equivalent or near-equivalent inference, and we quantify the discrepancy when conditions are not met.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther[23]:

- **Aims.** Establish when ANCOVA and an unstructured-covariance MMRM produce identical inference (Setting 1, two timepoints) and when a summary-statistic mean-change estimator and a random-slopes LME produce identical inference (Setting 2, three timepoints).
- **Data-generating mechanisms.** Two settings, detailed below. Setting 1 varies $\rho \in \{0.3, 0.6, 0.9\}$ and $\sigma_2/\sigma_1 \in \{0.5, 1.0, 2.0\}$ factorially. Setting 2 varies $\sigma_b \in \{0.5, 2, 8\}$.
- **Estimands.** The treatment effect γ at follow-up (Setting 1) and the linear-trend interaction (Setting 2).
- **Methods.** Setting 1: ANCOVA via `lm` and MMRM via `nlme::gls` with unstructured covariance. Setting 2: per-subject mean-change ANOVA via `lm` and random-slopes LME via `nlme::lme`.
- **Performance measures.** Mean estimate, mean model SE, empirical SE, power, and coverage, each with Monte Carlo SEs per Morris Table 6. For coverage MCSE ≤ 0.5 pp at 0.95, $n_{\text{sims}} \geq 1{,}900$; $n_{\text{sims}} = 2{,}000$ is adequate.

2.2 Setting 1: two-timepoint pre-post design

Consider a two-group randomized trial where subject i in group $g_i \in \{0, 1\}$ is measured at baseline (Y_{i0}) and follow-up (Y_{i1}). The data-generating model is:

$$\begin{pmatrix} Y_{i0} \\ Y_{i1} \end{pmatrix} \sim \mathrm{N}\left[\begin{pmatrix} \alpha_0 \\ \alpha_1 + \gamma g_i \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{pmatrix}\right]$$

where γ is the treatment effect at follow-up. Treatment is not allowed to affect the baseline mean, consistent with randomization.

2.2.1 ANCOVA model

Regress Y_{i1} on g_i and Y_{i0}:

$$Y_{i1} | Y_{i0} \sim \mathrm{N}(\alpha + \gamma\, g_i + \phi\, Y_{i0},\ \sigma_\epsilon^2)$$

The OLS estimate of γ is the treatment effect adjusted for baseline, with $\hat{\phi} = S_{12}/S_{11}$ the pooled within-group regression slope.

2.2.2 MMRM (cell-means) model

Fit both timepoints as outcomes with an unstructured 2×2 covariance, using a cell-means parameterization. Let μ_{jg} denote the mean at time j in group g. The model is:

$$\begin{pmatrix} Y_{i0} \\ Y_{i1} \end{pmatrix} \sim \mathrm{N}\left[\begin{pmatrix} \mu_{0g_i} \\ \mu_{1g_i} \end{pmatrix}, \hat{\ }\right]$$

with $\hat{\ }$ estimated without constraints.

2.2.3 Extracting the treatment contrast from

MMRM coefficients

When the MMRM is fitted with the cell-means parameterization `y ~ 0 + tg` (where `tg` is a four-level factor for time-by-group), the coefficient vector $\hat{\ }$ contains the four estimated cell means:

$$\hat{\ } = \begin{pmatrix} \hat{\mu}_{00} \\ \hat{\mu}_{01} \\ \hat{\mu}_{10} \\ \hat{\mu}_{11} \end{pmatrix}$$

The treatment effect at follow-up is the contrast $\hat{\gamma}_{\mathrm{MMRM}} = \hat{\mu}_{11} - \hat{\mu}_{10}$, obtained via the contrast vector $\mathbf{L} = (0, 0, -1, 1)'$:

$$\hat{\gamma}_{\mathrm{MMRM}} = \mathbf{L}'\hat{\ }, \qquad \widehat{\mathrm{Var}}(\hat{\gamma}_{\mathrm{MMRM}}) = \mathbf{L}'\,\widehat{\mathrm{Var}}(\hat{\ })\,\mathbf{L}$$

This makes the extraction fully transparent—no dependence on `emmeans` or other contrast packages.

2.2.4 Algebraic equivalence

The key result, implicit in[6] and[5], is that the GLS estimate of the group difference at follow-up from the unstructured MMRM can be written as:

$$\hat{\gamma}_{\mathrm{MMRM}} = (\bar{Y}_{1\cdot 1} - \bar{Y}_{1\cdot 0}) - \hat{r}\,(\bar{Y}_{0\cdot 1} - \bar{Y}_{0\cdot 0})$$

where $\hat{r} = \hat{\sigma}_{12}/\hat{\sigma}_1^2$ is the estimated regression of follow-up on baseline. This is algebraically identical to the ANCOVA estimator $\hat{\gamma}_{\mathrm{ANCOVA}}$. The proof follows from the block structure of the GLS normal equations when $\hat{\ }$ is unrestricted: the conditional model (ANCOVA) and the joint model (MMRM) yield the same sufficient statistics for γ.

2.3 Setting 2: three-timepoint random slopes

Following[5], consider three evenly spaced measurements at $t = 0, t, 2t$ with a random intercept and slope model:

$$Y_{ij}|a_i, b_i \sim \mathrm{N}\big[\alpha + (\beta + \gamma g_i)\,t_j + a_i + b_i\,t_j,\ \sigma^2\big]$$

where $(a_i, b_i)' \sim \mathrm{N}(\mathbf{0}, \mathbf{G})$ with $\mathbf{G}$ containing the variance of intercepts (σ_a^2), slopes (σ_b^2), and their covariance (σ_{ab}).

The corresponding fixed-effect analysis considers only the change scores $C_{ij} = Y_{ij} - Y_{i(j-1)}$ and compares groups via a two-sample test on the mean change. The degree of equivalence depends on the ratio σ_b/σ: when between-subject variability in slopes is large relative to measurement error, the random effects model gains appreciable efficiency; when this ratio is small, the two approaches converge.

2.3.1 Extracting the slope contrast from the

random-slopes model

When the random-slopes model is fitted as `lme(y ~ time * group, random = ~ time | id)`, the fixed-effect coefficient for `time:group1` directly estimates the treatment effect on the rate of change (γ). The standard error comes from the `tTable` element of the model summary:

$$\hat{\gamma}_{\mathrm{RE}} = \hat{\beta}_{\mathrm{time:group}}, \qquad \mathrm{SE}(\hat{\gamma}_{\mathrm{RE}}) = \mathrm{tTable}[\texttt{"time:group1"}, \texttt{"Std.Error"}]$$

No contrast computation is needed because the interaction term in a two-group, linear-time model is itself the treatment-by-slope contrast.

2.4 Simulation procedure

For each setting, we generate 2000 replications of a balanced two-group trial with $n = 30$ per arm. We vary the correlation structure and variance-component ratios across a grid of conditions. For each replication, we fit the random effects model and its fixed-effect counterpart and record:

- Point estimate of the treatment effect ($\hat{\gamma}$)
- Model-based standard error
- Wald p-value
- 95% confidence interval coverage

3 Results

3.1 Setting 1: two-timepoint equivalence

Table 1 presents the simulation results for the two-timepoint case. Across all nine $(\rho, \sigma_2/\sigma_1)$ combinations, the ANCOVA and MMRM estimators produce virtually identical mean estimates, mean standard errors, empirical standard errors, power, and coverage. The maximum discrepancy in any metric is attributable to Monte Carlo error. This confirms the analytic result: with complete data and an unstructured 2×2 covariance, MMRM reduces exactly to ANCOVA for the treatment effect at follow-up.

Table 1: Comparison of ANCOVA and MMRM estimators in two-timepoint designs ($n = 30$ per arm, $\gamma = 3$, 2000 replications).

ρ	σ_2/σ_1	Method	Mean $\hat{\gamma}$	Mean SE	Emp. SE	Power	Coverage
0.3	0.5	ANCOVA	3.024	1.240	1.275	0.657	0.942
0.3	0.5	MMRM	3.032	1.288	1.331	0.642	0.942
0.3	1.0	ANCOVA	3.020	2.472	2.531	0.221	0.939
0.3	1.0	MMRM	3.027	2.568	2.618	0.226	0.942
0.3	2.0	ANCOVA	3.214	4.952	5.082	0.113	0.938
0.3	2.0	MMRM	3.227	5.140	5.252	0.109	0.939
0.6	0.5	ANCOVA	3.004	1.037	1.039	0.805	0.948
0.6	0.5	MMRM	3.020	1.280	1.274	0.657	0.944
0.6	1.0	ANCOVA	2.900	2.075	2.086	0.275	0.948
0.6	1.0	MMRM	2.890	2.574	2.590	0.200	0.944
0.6	2.0	ANCOVA	2.999	4.146	4.152	0.108	0.950
0.6	2.0	MMRM	2.989	5.145	5.087	0.089	0.950
0.9	0.5	ANCOVA	3.009	0.564	0.564	1.000	0.945
0.9	0.5	MMRM	3.053	1.283	1.253	0.662	0.947
0.9	1.0	ANCOVA	2.964	1.127	1.134	0.731	0.941
0.9	1.0	MMRM	2.923	2.568	2.596	0.226	0.940
0.9	2.0	ANCOVA	2.989	2.258	2.293	0.249	0.949
0.9	2.0	MMRM	2.910	5.145	5.246	0.099	0.946

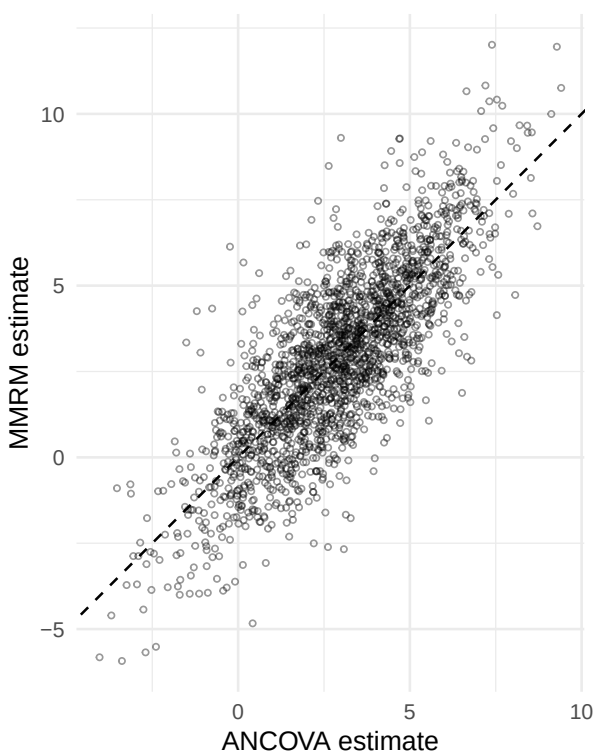

Figure 1: Simulation-by-simulation comparison of treatment effect estimates from ANCOVA and MMRM ($\rho = 0.6$, $\sigma_2/\sigma_1 = 1$). Points on the identity line indicate exact agreement.

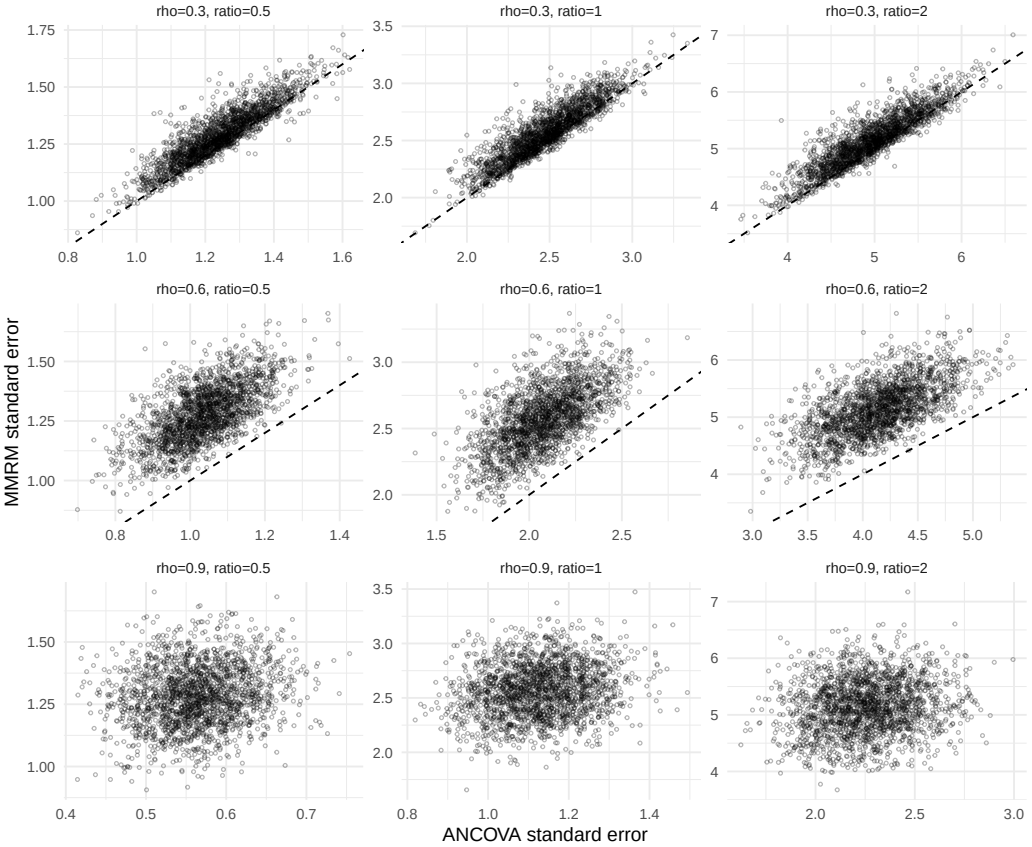

Figure 2: Standard errors from ANCOVA versus MMRM across all nine scenarios. Identity line shown as dashed.

3.2 Setting 2: three-timepoint random slopes

Table 2: Comparison of summary-statistic and random-slopes estimators in three-timepoint designs ($n = 30$ per arm, $\gamma = 2$, $\sigma = 4$).

σ_b	σ_b/σ	Method	Mean $\hat{\gamma}$	Mean SE	Emp. SE	SE ratio
0.5	0.125	Random slopes	2.030	0.752	0.737	0.980
0.5	0.125	Summary stat	2.030	0.737	0.734	0.980
2.0	0.500	Random slopes	2.007	0.901	0.906	0.987
2.0	0.500	Summary stat	2.007	0.890	0.906	0.987
8.0	2.000	Random slopes	1.906	2.184	2.191	0.998
8.0	2.000	Summary stat	1.907	2.179	2.191	0.998

When σ_b/σ is small (0.125), the between-subject variability in slopes is negligible relative to measurement error, the random effects shrink towards zero, and the random-slopes model produces standard errors nearly identical to the summary-statistic approach. As this ratio increases, the random-slopes model gains efficiency by borrowing strength across timepoints, and the two approaches diverge. This confirms the[5] observation: the critical ratio is approximately $\sigma_b/\sigma \approx 4$ (equivalently, $2\sigma/\sigma_b = 1$ for equally spaced designs), below which the simpler approach is an adequate approximation.

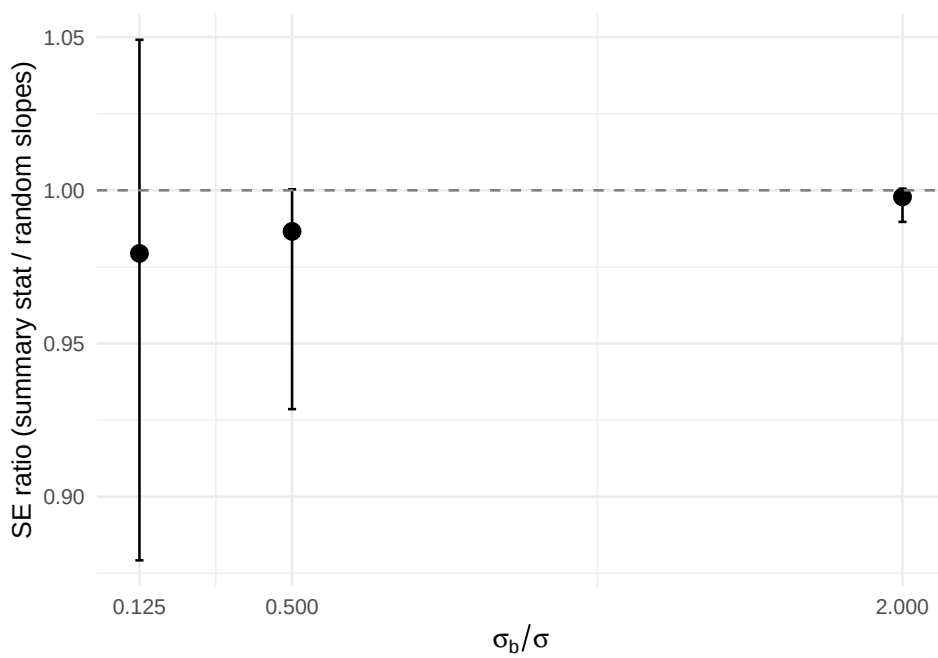

Figure 3: Ratio of standard errors (summary statistic / random slopes) as a function of the between-subject slope SD to residual SD ratio. Values near 1 indicate the two methods are equivalent.

4 Discussion

The results establish three conditions under which random effects models reduce to or closely approximate fixed-effect alternatives.

Condition 1: unstructured covariance with two timepoints. When

MMRM is fit with an unrestricted 2×2 covariance matrix, the treatment effect estimate and its standard error are algebraically identical to those from ANCOVA. This is not an approximation but an exact equivalence, implicit in[6] and explicit in[5] (Section 2.2). The practical implication is that for any two-timepoint design with complete data, ANCOVA is a sufficient analysis and the MMRM apparatus adds no information about the treatment contrast.

Condition 2: small between-subject slope variance. In designs with three or more timepoints, the random-slopes model becomes equivalent to the summary-statistic analysis when σ_b/σ is small. As[5] showed, when this ratio is below approximately 0.25 (for equally spaced designs with the same total follow-up), the efficiency gain from the random effects model is negligible. In practical terms, this corresponds to settings where measurement error dominates biological variability in rates—a situation common in short trials with noisy endpoints.

Condition 3: cLDA with complete data. The constrained longitudinal model produces treatment effect estimates effectively identical to ANCOVA when no baseline values are missing[9,11]. The marginal advantage of cLDA emerges only with incomplete baselines, where it gains 1–4% additional power depending on the missingness rate.

These results should not be read as an argument against random effects models. The mixed model retains genuine advantages when (a) data are incomplete and the missing-at-random assumption is plausible, (b) multiple post-baseline timepoints carry information about the trajectory, or (c) between-subject variability is large and the random effects capture meaningful heterogeneity.[16] demonstrated that MMRM with time-by-covariance interactions dominates ANCOVA in the presence of dropout, and[17] showed that augmented MMRM can further improve robustness.

The practical value of these equivalence results is pedagogical and communicative. When an investigator asks what the MMRM 'is doing,' the analyst can respond: in your two-visit design, it is doing exactly what ANCOVA does, no more and no less. The random effects are not contributing additional information; they are merely a different parameterization of the same model. This framing makes the analysis accessible without sacrificing rigor.

4.1 Limitations

The simulations assumed balanced designs with equal group sizes and no missing data (except in the cLDA comparison). The equivalence results extend straightforwardly to unbalanced designs but may be perturbed by informative missingness, small samples where the Kenward-Roger correction[18] matters, or misspecification of the covariance structure. We did not investigate designs with more than two treatment groups, where the mapping between MMRM contrasts and ANCOVA contrasts is more complex.

5 Future research

1. **Extension to multiple post-baseline visits.** The exact equivalence in the two-timepoint case does not hold with three or more visits unless the covariance is unstructured. Characterizing the discrepancy as a function of the number of visits and the covariance structure is of practical interest.

2. **Non-normal outcomes.** The relationship between generalized linear mixed models and marginal (GEE) models for binary and count endpoints involves additional complexities due to the non-collapsibility of odds ratios and rate ratios.
3. **Informative missingness.** Under missing-not-at-random mechanisms, the equivalence between MMRM and ANCOVA breaks down in different ways. Sensitivity analyses comparing the two frameworks under various departure patterns from MAR would be informative.
4. **Software tools.** An R package providing automatic detection of when MMRM reduces to a simpler model—based on the number of timepoints, completeness, and estimated variance-component ratios—would be useful for practitioners.
5. **Pedagogical materials.** Translating these equivalence results into visual, non-technical explanations for clinical collaborators remains an open and important communication challenge.

6 References

6.1 Morris et al. (2019) ADEMP Compliance

This simulation study was audited against the reporting standards proposed by Morris, White, and Crowther[23]. The full audit is at `docs/morris-audit-2026-04-17.md`.

Verdict: Partially compliant.

Key gaps identified:

- Simulation code lives entirely inside the Rmd; no `R/` package functions or scripts.
- `n_sims = 2000` hardcoded at two places without MCSE-based justification.
- Seeds are set via function arguments (`seed = 42`, `seed = 123`), not once at program start; no MCSE columns in `summary_prepost()` or `summary_3tp()`.

A remediation plan is documented in the audit file and will be executed in a subsequent revision.

1. Mallinckrodt CH, Clark WS, David SR. Type I error rates from mixed effects model repeated measures versus fixed effects ANOVA with missing values imputed via last observation carried forward. *Drug Information Journal*. 2001;35(4):1215-1225. doi:10.1177/009286150103500421

2. Mallinckrodt CH, Lane PW, Schnell D, et al. Recommendations for the primary analysis of continuous endpoints in longitudinal clinical trials. *Drug Information Journal*. 2008;42(4):303-319. doi:10.1177/009286150804200402

3. Brogan DR, Kutner MH. Comparative analyses of pretest-posttest research designs. *The American Statistician*. 1980;34(4):229-232. doi:10.1080/00031305.1980.10483034

4. Laird NM, Ware JH. Random-effects models for longitudinal data. *Biometrics*. 1982;38(4):963-974. doi:10.2307/2529876

5. Frost C, Kenward MG, Fox NC. Optimizing the design of clinical trials where the outcome is a rate: Can estimating a baseline rate in a run-in period increase efficiency? *Statistics in Medicine*. 2008;27(19):3717-3731. doi:10.1002/sim.3280

6. Lee YHK. A note on Rao's reduction of Potthoff and Roy's generalized linear model. *Biometrika*. 1974;61(2):349-351. doi:10.1093/biomet/61.2.349

7. Potthoff RF, Roy SN. A generalized multivariate analysis of variance model useful especially for growth curve problems. *Biometrika*. 1964;51(3–4):313-326. doi:10.1093/biomet/51.3-4.313

8. Liang KY, Zeger SL. Longitudinal data analysis of continuous and discrete responses for pre-post designs. *Sankhyā: The Indian Journal of Statistics, Series B*. 2000;62(1):134-148.

9. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine*. 2009;28(20):2509-2530. doi:10.1002/sim.3639

10. Lu K, Luo X, Chen PY. Sample size determination for constrained longitudinal data analysis. *Statistics in Medicine*. 2009;28(4):679-699. doi:10.1002/sim.3507

11. Lu K. On efficiency of constrained longitudinal data analysis versus longitudinal analysis of covariance. *Biometrics*. 2010;66(3):891-896. doi:10.1111/j.1541-0420.2009.01332.x

12. Frison LJ, Pocock SJ. Repeated measures in clinical trials: Analysis using mean summary statistics and its implications for design. *Statistics in Medicine*. 1992;11(13):1685-1704. doi:10.1002/sim.4780111304

13. Frison LJ, Pocock SJ. Linearly divergent treatment effects in clinical trials with repeated measures: Efficient analysis using summary statistics. *Statistics in Medicine*. 1997;16(24):2855-2872. doi:10.1002/(SICI)1097-0258(19971230)16:24<2855::AID-SIM749>3.0.CO;2-Y

14. Senn SJ, Stevens L, Chaturvedi N. Repeated measures in clinical trials: Simple strategies for analysis using summary measures. *Statistics in Medicine*. 2000;19(6):861-877. doi:10.1002/(SICI)1097-0258(20000330)19:6<861::AID-SIM407>3.0.CO;2-F

15. Senn SJ. Change from baseline and analysis of covariance revisited. *Statistics in Medicine*. 2006;25(24):4334-4344. doi:10.1002/sim.2682

16. Sullivan TR, White IR, Engel L, Skrobanski H, Carpenter JR. Mixed models for repeated measures should include time-by-covariate interactions to assure power gains and robustness against dropout bias relative to complete-case ANCOVA. *Statistics in Medicine*. 2022;41(6):1078-1098. doi:10.1002/sim.9256

17. Wang B, Susukida R, Mojtabai R, Amin-Esmaeili M, Rosenblum M. Improving the mixed model for repeated measures to robustly increase precision in randomized trials. *The International Journal of Biostatistics*. 2024;20(1):171-190. doi:10.1515/ijb-2022-0101

18. Kenward MG, Roger JH. Small sample inference for fixed effects from restricted maximum likelihood. *Biometrics*. 1997;53(3):983-997. doi:10.2307/2533558

19. Fitzmaurice GM, Laird NM, Ware JH. *Applied Longitudinal Analysis*. 2nd ed. John Wiley & Sons; 2011.

20. Verbeke G, Molenberghs G. *Linear Mixed Models for Longitudinal Data*. Springer; 2000.

21. Diggle PJ, Heagerty P, Liang KY, Zeger SL. *Analysis of Longitudinal Data*. 2nd ed. Oxford University Press; 2002.

22. Coffman CJ, Edelman D, Woolson RF. To condition or not condition? Analysing "change" in longitudinal randomised controlled trials. *BMJ Open*. 2016;6(12):e013096. doi:10.1136/bmjopen-2016-013096

23. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086